

AI, Data Centers, Water, Energy, and Everyday Digital Life

An integrated environmental comparison report for 2025 and 2026

Prepared with current public sources and explicit estimates

May 1, 2026

This report uses an inverted-pyramid structure: the highest-value conclusions and factoids come first, followed by integrated comparisons, then methods and source notes.

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Lead: the most important finding

The most important distinction is between individual use and system-scale buildout. One ordinary hour of LLM chat is usually tiny compared with one hour of driving, high-end gaming, or air conditioning. But the aggregate AI/data-center buildout is large, fast-growing, locally concentrated, and often poorly disclosed.

The clearest numerical summary is this:

- Global data centers used about 486 TWh of electricity in 2025, based on the International Energy Agency's 2024 estimate of 415 TWh and its reported 17% rise in 2025. That is already larger than the annual electricity consumption of many countries. Source: International Energy Agency, "Energy and AI" and April 2026 news release, <https://www.iea.org/reports/energy-and-ai/energy-demand-from-ai> and <https://www.iea.org/news/data-centre-electricity-use-surge-in-2025-even-with-tightening-bottlenecks-driving-a-scramble-for-solutions>.
- Direct data-center cooling water was probably much lower than U.S. golf-course irrigation in 2025, perhaps 45-90 billion gallons globally for data centers versus about 531 billion gallons for U.S. golf courses. But when indirect water used to generate data-center electricity is included, the data-center water footprint may be comparable to U.S. golf-course irrigation. Sources: Lawrence Berkeley National Laboratory, "2024 United States Data Center Energy Usage Report," https://eta-publications.lbl.gov/sites/default/files/2024-12/lbnl-2024-united-states-data-center-energy-usage-report_1.pdf; GCSAA, "Golf Course Environmental Profile - Water Use and Management Practices," <https://www.gcsaa.org/docs/default-source/what-we-do/gcep-phase-4-water-report.pdf>.
- AI-specific electricity use in 2025 was probably in the same broad range as global video-game electricity use, and may already have exceeded it. A defensible range is roughly 100-240 TWh for AI systems and 120-200 TWh for the global video-game ecosystem. The AI estimate is uncertain because companies do not consistently disclose AI-specific electricity use. Sources: IEA; de Vries-Gao, "The carbon and water footprints of data centers and what this could mean for artificial intelligence," Patterns, <https://www.sciencedirect.com/science/article/pii/S2666389925002788>; Lawrence Berkeley National Laboratory, "Taming the Energy Use of Gaming Computers," <https://datacenters.lbl.gov/publications/taming-energy-use-gaming-computers>.
- One hour of gasoline-car driving is usually hundreds to thousands of times more carbon-intensive than one hour of ordinary LLM chat. EPA estimates the average passenger vehicle emits about 400 g CO₂ per mile; Epoch AI estimates a typical GPT-4o-style query at about 0.3 Wh. Sources: U.S. EPA, <https://www.epa.gov/greenvehicles/greenhouse-gas-emissions-typical-passenger-vehicle>; Epoch AI, "How much energy does ChatGPT use?" <https://epoch.ai/gradient-updates/how-much-energy-does-chatgpt-use>.

The practical conclusion: prompt-level guilt is usually the wrong frame; infrastructure accountability is the right frame. The questions that matter most are where data centers are built, what powers them, how they are cooled, whose water and grid capacity they use, whether fossil generation is extended or added, and how transparently companies report those impacts.

At-a-glance estimates

These figures are intended as defensible order-of-magnitude comparisons, not exact accounting.

Metric ; 2025 estimate ; 2026 extrapolation ; Confidence

Global data-center electricity, all uses ; ~486 TWh ; ~550-570 TWh ; Medium-high

AI-specific electricity ; ~100-240 TWh ; ~140-320 TWh ; Low-medium

Global video-game ecosystem electricity ; ~120-200 TWh ; ~125-210 TWh ; Low-medium

Bitcoin mining electricity ; ~138 TWh, annualized 2024/2025 anchor ; ~130-220 TWh, price-dependent ; Medium-low

Global EV charging electricity ; ~250-300 TWh if 2024's 180 TWh grew strongly ; ~330-420 TWh ; Medium-low

Global data-center direct water ; ~45-90 billion gallons ; ~55-110 billion gallons ; Medium-low

Global data-center water incl. electricity generation ; ~500-700+ billion gallons ; ~600-800+ billion gallons ; Low-medium

U.S. golf-course irrigation ; ~531 billion gallons ; ~500-560 billion gallons ; Medium-high

U.S. residential outdoor water use ; ~2.9-3.3 trillion gallons/year ; similar ; Medium

One hour gasoline driving ; ~14-18 kg CO₂e ; similar ; Medium

One hour ordinary LLM chat ; ~5-200 g CO₂e; more for heavy reasoning ; variable ; Low-medium

One hour video streaming ; ~36-56 g CO₂e in cited 2019/2020 European estimates ; likely lower on cleaner grids, higher on TVs/dirty grids ; Medium

One hour high-end PC gaming ; often hundreds of Wh; emissions depend on grid ; similar or modestly lower per device ; Medium-low

Ranked findings: what matters most

1. The biggest risk is infrastructure scale, not one prompt

An ordinary LLM query is likely small. Epoch AI estimated a typical GPT-4o-style text query at roughly 0.3 Wh, with long-context and reasoning queries potentially much higher. Even 200 ordinary queries in an hour would be only about 60 Wh before broader uncertainty factors. Source: Epoch AI, "How much energy does ChatGPT use?" <https://epoch.ai/gradient-updates/how-much-energy-does-chatgpt-use>.

By contrast, global data centers are now an infrastructure-scale electricity load. IEA's reported 17% rise in 2025 implies about 486 TWh for the year, and its base case projects about 945 TWh by 2030. Source: IEA, "Energy and AI," <https://www.iea.org/reports/energy-and-ai/energy-demand-from-ai>.

2. Water comparisons depend completely on boundaries

If the question is direct water used onsite for cooling, U.S. golf courses probably use much more water than all global data centers. If the question includes indirect water consumed to generate data-center electricity, global data centers may be roughly comparable to U.S. golf-course irrigation.

This is not a technicality. A water number without a boundary - direct consumption, withdrawal, potable use, non-potable use, indirect electricity water, or full lifecycle water - can mislead.

3. AI and gaming are now comparable electricity loads, but AI is growing faster

Gaming is a large and under-measured digital load. Lawrence Berkeley estimated gaming PCs alone used 75 TWh/year globally in 2012, before today's larger gaming market. But AI-focused data-center demand is growing faster than ordinary electricity demand, and IEA says AI-focused data centers grew even faster than the 17% overall data-center increase in 2025. Sources: LBNL, <https://datacenters.lbl.gov/publications/taming-energy-use-gaming-computers>; IEA April 2026 news release, <https://www.iea.org/news/data-centre-electricity-use-surge-in-2025-even-with-tightening-bottlenecks-driving-a-scramble-for-solutions>.

4. The most useful new comparisons are with streaming, gaming, crypto, EV charging, lawns, and renewable buildout

These comparisons place AI in the world people already understand: cars, lawns, golf courses, streaming, games, Bitcoin, EVs, household electricity, and the clean-energy buildout.

Integrated comparison analysis

A. Individual activity comparisons

A1. One hour of gasoline driving vs. one hour of LLM use

EPA estimates a typical passenger vehicle emits about 400 g CO₂ per mile and 8,887 g CO₂ per gallon of gasoline burned. At 30 miles per hour, one hour of driving produces about 12 kg tailpipe CO₂. Adding upstream fuel production, refining, and distribution gives a reasonable all-in estimate of 14-18 kg CO₂e per hour. Source: U.S. EPA, "Greenhouse Gas Emissions from a Typical Passenger Vehicle," <https://www.epa.gov/greenvehicles/greenhouse-gas-emissions-typical-passenger-vehicle>.

For LLM use, Epoch AI estimates a typical GPT-4o-style query at about 0.3 Wh. An hour of ordinary chat might involve 20-200 turns, or roughly 6-60 Wh before adding data-center overhead and grid emissions. A heavier reasoning or long-document hour could be substantially higher, perhaps 0.1-0.5+ kWh. Source: Epoch AI, <https://epoch.ai/gradient-updates/how-much-energy-does-chatgpt-use>.

Conclusion: one hour of gasoline driving is usually hundreds to thousands of times more carbon-intensive than one hour of ordinary LLM use.

A2. One hour of LLM use vs. one hour of video streaming

Carbon Trust estimated the carbon impact of one hour of video-on-demand streaming in Europe at about 55-56 g CO₂e/hour in 2020, with energy use around 188 Wh/hour and the viewing device responsible for the largest share. IEA's central estimate for one hour of streaming video in 2019 was 36 g CO₂, after correcting earlier inflated claims. Sources: Carbon Trust, "Carbon impact of video streaming," <https://www.carbontrust.com/our-work-and-impact/guides-reports-and-tools/carbon-impact-of-video-streaming>; IEA, "The carbon footprint of streaming video: fact-checking the headlines," <https://www.iea.org/commentaries/the-carbon-footprint-of-streaming-video-fact-checking-the-headlines>.

Ordinary LLM chat could be lower than video streaming if the user is asking short questions and reading text. Heavy LLM use with long documents, reasoning models, code execution, or image/video generation can exceed ordinary streaming.

Conclusion: ordinary text LLM use is often in the same general personal-use range as, or lower than, video streaming; heavy AI workflows are a different category.

A3. One hour of LLM use vs. one hour of high-end PC gaming

High-end PC gaming can draw hundreds of watts at the user device alone, before considering monitors, networking, game servers, or manufacturing. Lawrence Berkeley's study estimated a typical gaming computer

including display at about 1,400 kWh/year, equivalent to ten game consoles, six standard PCs, or three refrigerators; the more intensive segments could use double that. Source: LBNL, "Taming the Energy Use of Gaming Computers," <https://datacenters.lbl.gov/publications/taming-energy-use-gaming-computers>.

A gaming rig drawing 300-700 W for one hour uses 0.3-0.7 kWh at home. That is often more electricity than an hour of ordinary LLM chat, though not necessarily more than heavy long-context or agentic AI use.

Conclusion: on a per-user-hour basis, high-end gaming can easily exceed ordinary LLM chat. On an economy-wide basis, AI is growing faster and is more concentrated in grid-scale facilities.

A4. One LLM query vs. one web search, page load, or email attachment

This is a useful comparison, but current public data are weak because search engines and AI systems do not publish consistent per-query energy figures. Epoch AI notes that a commonly cited old claim treated a ChatGPT query as about 3 Wh, about ten times a Google search, but Epoch argues the updated GPT-4o-style figure is closer to 0.3 Wh for a typical text query. Source: Epoch AI, <https://epoch.ai/gradient-updates/how-much-energy-does-chatgpt-use>.

The better conclusion is not that one query equals one search. It is that the range of LLM queries is wide. A short answer, a long legal memo, a code-agent run, and a document-heavy reasoning task are not environmentally equivalent.

Conclusion: this comparison should be included in the report mainly to warn against false precision. Query energy should be reported by workload type: short text, long text, reasoning, document upload, image generation, video generation, and agentic automation.

B. Annual digital-sector comparisons

B1. AI electricity vs. video-game electricity

A reasonable 2025 range for AI-specific electricity is 100-240 TWh, based on IEA's global data-center total and studies estimating AI's share of data-center demand. De Vries-Gao estimated that AI applications accounted for 15-20% of global data-center electricity in 2024 and that AI systems could require up to about 23 GW in 2025, equivalent to about 201 TWh if sustained for a year. Sources: IEA; VU Amsterdam summary of de Vries-Gao research, <https://vu.nl/en/news/2025/ai-s-hidden-carbon-and-water-footprint>; Patterns article, <https://www.sciencedirect.com/science/article/pii/S2666389925002788>.

A reasonable 2025 range for the global video-game ecosystem is 120-200 TWh, combining older PC-gaming estimates, consoles, displays, mobile gaming, networks, cloud gaming, and industry operations. This is less certain than the data-center total because the gaming sector lacks a single authoritative annual global tracker.

Conclusion: AI and gaming were probably similar electricity-scale sectors in 2025. AI likely becomes clearly larger in 2026 under most reasonable growth assumptions.

B2. AI/data centers vs. cryptocurrency mining

IEA estimated cryptocurrency mining electricity use at about 110 TWh in 2022. Cambridge's 2025 Digital Mining Industry Report estimated Bitcoin's annual electricity consumption at about 138 TWh, or about 0.5% of global consumption, with network-wide emissions around 39.8 Mt CO₂e. Sources: IEA data centers page, <https://www.iea.org/energy-system/buildings/data-centres-and-data-transmission-networks>; Cambridge Judge

Business School, "Cambridge study: sustainable energy rising in Bitcoin mining,"

<https://www.jbs.cam.ac.uk/2025/cambridge-study-sustainable-energy-rising-in-bitcoin-mining/>.

If AI used roughly 100-240 TWh in 2025, then AI-specific electricity was probably already comparable to, and possibly greater than, Bitcoin mining.

Conclusion: crypto is a useful benchmark because it shows how fast a specialized compute load can become a public grid issue. AI differs because it is more integrated into mainstream cloud infrastructure and has a broader set of social and economic uses.

B3. Data-center electricity growth vs. EV charging growth

IEA's Global EV Outlook 2025 says the global EV fleet consumed about 180 TWh of electricity in 2024, almost 60% more than the previous year. Source: IEA, "Global EV Outlook 2025 - Outlook for energy demand,"

<https://www.iea.org/reports/global-ev-outlook-2025/outlook-for-energy-demand>.

If EV charging grew strongly again in 2025, a plausible 2025 range is about 250-300 TWh, still below total global data-center electricity of about 486 TWh. However, EV electricity substitutes for gasoline and diesel, while data-center electricity usually represents new service demand.

Conclusion: EVs and data centers are both major new grid loads. The climate interpretation differs: EV charging can displace oil combustion; AI/data-center load must be judged by what it enables, what powers it, and whether it delays grid decarbonization.

B4. Data centers vs. U.S. household electricity

EIA reports that the average U.S. household consumes about 10,500 kWh/year of electricity. Source: U.S. EIA, "Electricity use in homes," <https://www.eia.gov/energyexplained/use-of-energy/electricity-use-in-homes.php>.

Using that benchmark:

- Global data centers in 2025: 486 TWh = 486 billion kWh = electricity use of about 46 million average U.S. households.
- AI-specific electricity at 100-240 TWh = electricity use of about 10-23 million average U.S. households.
- Bitcoin at 138 TWh = electricity use of about 13 million average U.S. households.

Conclusion: household equivalents make the scale legible, but they should not imply that household use and data-center use have the same social purpose or grid behavior.

C. Water and land-use comparisons

C1. Global data-center water vs. U.S. golf-course water

GCSAA reports U.S. golf facilities applied 1.63 million acre-feet of water in 2024, equivalent to about 531 billion gallons. Source: GCSAA, <https://www.gcsaa.org/docs/default-source/what-we-do/gc-ep-phase-4-water-report.pdf>.

LBNL estimated U.S. data centers in 2023 used 66 billion liters of direct onsite water and nearly 800 billion liters of indirect water through electricity generation. Scaling LBNL's U.S. water intensities to the estimated global 2025 data-center electricity total yields about 48 billion gallons direct and about 584 billion gallons including indirect power-generation water. Source: LBNL, https://eta-publications.lbl.gov/sites/default/files/2024-12/lbnl-2024-united-states-data-center-energy-usage-report_1.pdf.

Conclusion: direct cooling water is much lower than U.S. golf-course irrigation; broader electricity-linked water may be comparable.

C2. Data-center water vs. U.S. residential lawns and landscaping

EPA WaterSense says residential outdoor water use in the United States accounts for nearly 8-9 billion gallons per day, mainly for landscape irrigation. That equals about 2.9-3.3 trillion gallons per year. Sources: EPA WaterSense Outdoors, <https://www.epa.gov/watersense/outdoors>; EPA WaterSense Start Saving, <https://www.epa.gov/watersense/start-saving>.

Compared with that:

- U.S. golf-course irrigation: ~531 billion gallons/year, about one-sixth of U.S. residential outdoor water use.
- Global data-center direct cooling water: ~45-90 billion gallons/year, much lower than U.S. residential outdoor water use.
- Global data-center broader operational water: ~500-700+ billion gallons/year, still below residential outdoor water use but no longer trivial.

Conclusion: lawns and landscaping are an important comparison because they reveal how large everyday aesthetic water use is. But local water stress matters more than national totals.

C3. Data-center water vs. local municipal water capacity

A global annual number can conceal a local capacity problem. IEA notes that data centers in Ireland accounted for 18% of national electricity use in 2022, and data centers plus other non-industrial large energy users could account for 28% of Ireland's demand by 2031 without added generation capacity. Source: IEA, data centers and data transmission networks, <https://www.iea.org/energy-system/buildings/data-centre-s-and-data-transmission-networks>.

A 2026 preprint by Han, Li, Wierman, and Ren argues that U.S. data-center growth could require hundreds of millions of gallons per day of new water capacity through 2030, with impacts concentrated in host communities. This is emerging research rather than an official government estimate, but it highlights the right question: not only annual gallons, but peak water capacity on hot days. Source: Han et al., "Small Bottle, Big Pipe," arXiv, <https://arxiv.org/abs/2603.02705>.

Conclusion: local capacity - power, water, wastewater, transmission, and cooling constraints - is more policy-relevant than global totals.

D. Hardware, training, and lifecycle comparisons

D1. Training one large model vs. annual emissions of institutions or towns

Luccioni, Viguier, and Ligozat estimated that training BLOOM, a 176-billion-parameter model, emitted 24.7 tonnes CO₂e for dynamic power consumption and 50.5 tonnes CO₂e including broader processes from equipment manufacturing to energy-based operational consumption. Source: Luccioni et al., "Estimating the Carbon Footprint of BLOOM," Journal of Machine Learning Research, <https://www.jmlr.org/papers/volume24/23-0069/23-0069.pdf>.

Older estimates for GPT-3 training are much higher, commonly cited around 552 tonnes CO₂e, but they are sensitive to assumptions about hardware, utilization, data-center efficiency, and grid carbon intensity. Source: Patterson et al., "Carbon Emissions and Large Neural Network Training," <https://arxiv.org/abs/2104.10350>.

Conclusion: training emissions are vivid but no longer the whole story. As models are used by millions of people, inference and infrastructure growth can dominate over a single training run.

D2. Embodied carbon of GPUs and servers vs. operational electricity

The BLOOM lifecycle study estimated embodied emissions associated with BLOOM training at about 11.2 tonnes CO₂e, less than the 24.7 tonnes from dynamic training power and about 22% of the 50.5-tonne broader lifecycle estimate. Source: Luccioni et al., <https://www.jmlr.org/papers/volume24/23-0069/23-0069.pdf>.

This does not mean embodied carbon is unimportant. Rapid hardware turnover, chip fabrication, server manufacturing, data-center construction, diesel generators, and grid interconnections can become major uncounted impacts. But in the BLOOM case, operational energy was larger than the hardware manufacturing share.

Conclusion: a full report should include embodied carbon, but it should not assume hardware always dominates. The ratio depends on utilization, lifetime, grid mix, and replacement cycles.

D3. AI demand vs. renewable-energy additions and grid buildout

IEA's Electricity 2025 report says low-emission sources - renewables and nuclear - are expected to meet all global demand growth through 2027, with renewables meeting about 95% of the growth. It also says renewables are expected to overtake coal in 2025, and solar PV is expected to provide roughly half of global electricity-demand growth through 2027. Source: IEA, "Electricity 2025," executive summary and supply sections, <https://www.iea.org/reports/electricity-2025/executive-summary> and <https://www.iea.org/reports/electricity-2025/supply>.

This is encouraging globally, but it does not guarantee that every new data center is powered by new clean electricity at the right place and time. Grid congestion, interconnection queues, local gas generation, backup diesel generation, and delayed transmission can still make a specific data-center cluster environmentally and politically contentious.

Conclusion: the key policy comparison is not just "AI electricity vs. renewable electricity." It is whether new AI load is matched by additional, local, time-aligned clean power and transmission.

2026 extrapolations

These extrapolations use simple growth assumptions and should be read as scenario estimates.

Data centers and AI

IEA's base-case trajectory implies continued mid-teens annual growth in data-center electricity demand. If 2025 was about 486 TWh, then 2026 is plausibly 550-570 TWh.

AI-specific electricity could plausibly rise from 100-240 TWh in 2025 to 140-320 TWh in 2026, assuming AI-focused demand grows faster than the overall data-center category but is constrained by chips, power availability, permitting, and interconnection.

Gaming

Gaming electricity probably grows much more slowly than AI. A reasonable 2026 range is 125-210 TWh, unless cloud gaming, high-refresh displays, or gaming-PC sales accelerate strongly.

Crypto

Bitcoin electricity depends strongly on price, mining rewards, hardware efficiency, and regulation. Cambridge's 138 TWh anchor suggests a broad 2026 range of 130-220 TWh is defensible, but less stable than the data-center estimate.

EV charging

IEA reported 180 TWh of global EV electricity in 2024, almost 60% above the previous year. If growth remains strong, 2026 could plausibly reach 330-420 TWh. This could begin to approach global data-center electricity, but the climate interpretation differs because EVs replace liquid fuel combustion.

Water

If global data-center electricity rises to around 550-570 TWh in 2026 and water intensities remain similar:

- Direct data-center cooling water: 55-110 billion gallons.
- Broader water footprint including electricity generation: 600-800+ billion gallons.
- U.S. golf-course irrigation: likely 500-560 billion gallons, mostly weather-dependent.
- U.S. residential outdoor use: likely 2.9-3.3 trillion gallons, depending on weather and restrictions.

Methods and assumptions

Electricity totals

The global data-center figure is anchored to IEA's 415 TWh estimate for 2024 and its reported 17% increase in 2025:

$$415 \text{ TWh} \times 1.17 = 486 \text{ TWh.}$$

The 2026 estimate assumes roughly 13-17% additional growth, consistent with IEA's statement that data-center electricity consumption grows about 15% per year from 2024 to 2030 in its base case.

AI electricity

AI electricity is estimated as a share of global data-center electricity. Because companies do not consistently disclose AI-specific electricity, the report uses a range rather than a single number. The 2025 range of 100-240 TWh is consistent with AI representing roughly 20-50% of the 2025 global data-center total, supported by de Vries-Gao's estimate that AI applications were 15-20% of data-center electricity in 2024 and could reach around 23 GW of power demand in 2025.

Data-center water

LBNL estimated U.S. data centers consumed 176 TWh and 66 billion liters of direct water in 2023. That implies:

66 billion liters / 176 billion kWh = 0.375 liters/kWh direct water.

Applied to global 2025 data-center electricity:

486 billion kWh x 0.375 liters/kWh = 182 billion liters = about 48 billion gallons.

For indirect electricity-generation water, LBNL's 800 billion liters / 176 billion kWh implies about 4.55 liters/kWh:

486 billion kWh x 4.55 liters/kWh = 2.21 trillion liters = about 584 billion gallons.

Ranges are wider than these point estimates because global cooling systems, climate, grid mix, reporting boundaries, and water accounting differ substantially.

Golf-course water

GCSAA reports 1.63 million acre-feet in 2024. Conversion:

1 acre-foot = 325,851 gallons.

1.63 million acre-feet x 325,851 gallons = about 531 billion gallons.

The 2025 estimate assumes similar national use, with weather-driven variation.

Residential outdoor water

EPA WaterSense gives 8-9 billion gallons per day for residential outdoor water use. Annualized:

8 billion gallons/day x 365 = 2.92 trillion gallons/year.

9 billion gallons/day x 365 = 3.29 trillion gallons/year.

Driving emissions

EPA's 400 g CO₂/mile figure gives 12 kg CO₂ for 30 miles. The report adds an approximate 15-30% upstream fuel factor for production, refining, and distribution, producing a broad estimate of 14-18 kg CO₂e/hour.

LLM-use emissions

Epoch AI's 0.3 Wh/query estimate is multiplied by a range of possible query counts and workload types. The report treats ordinary text chat, long-context chat, reasoning models, image generation, and agentic workflows as different workload classes.

Source notes and reliability

The strongest sources in this report are IEA, LBNL, EPA, EIA, GCSAA, Carbon Trust, Cambridge Judge Business School/CCAF, and peer-reviewed or preprint academic work where noted. The weakest estimates are AI-specific electricity and global gaming electricity because both categories lack consistent annual public reporting.

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